

Product Requirement Document

Product Name

Inventory and Sales management dashboard (POS System)

Purpose

Provide a web-based system to manage products, inventory, suppliers, customers, sales/invoices and analytics for retail or small-to-medium businesses.

Problem Statement

Businesses lack a single integrated system to track inventory, create and email invoices, monitor sales trends, and manage users with role-based access while keeping APIs responsive under load.

Objective

Web-based POS dashboard that enables:

1. Real-time inventory tracking with automated low-stock alerts
2. Fast and accurate sales transaction processing with multiple payment methods
3. Comprehensive business analytics with trend analysis
4. Role-based user access and permissions management
5. Complete audit trail through stock movement logging
6. Data export for reporting and compliance (CSV, PDF)
7. Multi-user concurrent access with session management

Target Users

Primary: Store owners, inventory managers, sales staff

Secondary: Administrators and accountants

User role	Description	Primary tasks
Admin	Business owner/Operations	Full system access, user management, all

	manager	CRUD operations
Manager	Department/store manager	Sales processing, customer/supplier management, reporting
Staff	Salesman/cashier	Sales creation, customer lookup, product search, stock inquiry

Key features

The key features of the application are:

1. Dashboard & Analytics

- Summary Cards: Revenue, profit, customer count, refunds
- Revenue & Profit Trends: Period-based comparison (Today, 3 Months, 12 Months, Custom)
- Payment Method Analysis: Revenue breakdown by payment type
- Top Products: Best-selling items ranked by revenue
- Top Customers: Most valuable customers ranked by spending
- Top Categories: Category performance metrics
- Low Stock Alerts: Visual indicator of low inventory items
- Trend Indicators: % change and direction (increasing/decreasing/stable)

2. Product Management

- Centralized product catalog with SKU tracking
- Advanced filtering (price range, stock level, category, supplier)
- Search functionality (name, SKU, description)
- Per-product sales charts and performance metrics
- Cost price vs. selling price tracking
- Low stock limit customization
- Infinite scroll list pagination

3. Sales Management

- Point-of-Sale transaction creation
- Multiple payment methods (Cash, Card, UPI, Wallet, Net Banking)
- Discount application per transaction
- Customer selection and linking
- Line-item product selection with quantity
- Invoice generation with unique invoice numbers
- PDF invoice download
- Payment status tracking (Pending, Completed, Refunded)
- CSV export of sales data
- Sales history and detail view

4. Customer Management

- Customer database with contact information
- Customer detail pages showing order history
- Phone number and address tracking
- Creation/update timestamp audit trail
- Customer-linked sales history

5. Supplier Management

- Supplier database with contact details
- Supplier-to-product relationship tracking
- Supplier detail pages with linked products
- Email and phone number records
- Supplier information CRUD operations

6. Inventory & Stock Movement

- Complete stock movement audit log
- Movement type tracking (IN/OUT)
- Movement reason categorization (Purchase, Sale, Return, Adjustment, Transfer)
- User attribution (who logged the movement)
- Date-stamped entries
- Filter by product, movement type, reason, user, date range
- CSV export of stock movement reports

7. Categories

- Product category management
- Category-to-product linking
- Category detail pages

8. User Management (Admin Only)

- User account creation and management
- Role assignment (Admin, Manager, Staff)
- Username, email, first/last name tracking
- Superuser/Staff flag management
- Account creation timestamps

9. Authentication & Authorization

- Username/password-based login with JWT tokens
- HttpOnly cookie-based session storage
- User registration with role selection
- Forgot password flow with OTP verification
- Password reset functionality
- Session persistence across page reloads
- Automatic logout on token expiration
- Role-based route protection

10. UI/UX Features

- Dark mode / Light mode toggle
- Responsive design (mobile, tablet, desktop)
- Collapsible sidebar navigation
- Toast notifications for user feedback
- Skeleton loading states
- Infinite scroll lazy loading
- Smooth animations and transitions

User roles and permissions

Admin	Manager	Staff
View dashboard	View dashboard	View dashboard (Read only)
Create/edit/delete Products	Create/edit/delete Products	View Products (Search/Filter)
Create/edit/delete Sales	Create/edit/delete Sales	Create sales (creation only)
Create/edit/delete Customers	Create/edit/delete Customers	Edit/delete Sales
Create/edit/delete Suppliers	Create/edit/delete Suppliers	View customers (search/filter)
Create/edit/delete Categories	Create/edit/delete Categories	View stock movement log (read-only)
View stock movement log	View stock movement log	Create/edit/delete any master data
Create/edit/delete Users	Create/edit/delete Users	Create/edit/delete Users
Manage user roles	View user management page	View user management page
Access user management page		
Export Data (CSV, PDF)	Export Data (CSV, PDF)	
View all reports	View all reports	

Functional requirements

1. Authentication & Session Management

- User should login with username and password
- System should validate credentials against backend API
- System should store JWT token in HttpOnly cookie upon successful login
- System should maintain session across page reloads and browser tabs

- System should automatically redirect unauthenticated users to /sign-in
- System should prevent authenticated users from accessing /sign-in, /sign-up, /forgot-password
- System should provide forgot password functionality with OTP verification
- System should allow password reset after OTP verification
- System should support user registration with role selection (Admin/Manager/Staff)
- System should enforce role-based access control (RBAC) on routes
- System should prevent non-admin users from accessing /users page
- System should provide logout functionality that clears session

2. Dashboard

- Dashboard should display summary cards (Revenue, Profit, Customers, Refunds)
- Summary cards should show current period values and trend % change
- Dashboard should display revenue and profit line chart
- Chart should support period filtering (Today, 3M, 12M, Custom)
- Dashboard should display payment method revenue breakdown
- Dashboard should display top 5 selling products with revenue and quantity
- Dashboard should display top 5 customers with order count and total spent
- Dashboard should display low stock alert items
- Dashboard should show trend direction (increasing/decreasing/stable) with icon
- Dashboard should be filterable by time period with visual feedback

3. Product Management

- System should display paginated list of all products (limit: 10–50 items)
- Product list should support infinite scroll for lazy loading
- User should search products by name, SKU, or description
- User should filter products by category and supplier
- User should filter products by price range (cost and selling price)
- User should filter products by stock level (min/max)
- User should filter products by date range (created_after/before)

- User should apply low stock filter to show only items below limit
- Admin/Manager should create new products with all required fields
- Admin/Manager should edit existing product details
- Admin/Manager should delete products from catalog
- Product detail page should display per-product sales chart
- Product detail page should show cost vs. selling price comparison
- Product list should display low stock indicator (visual warning)
- System should validate product SKU uniqueness

4. Sales Management

- User shall view paginated list of all sales transactions
- Sales list shall support infinite scroll pagination
- User shall search sales by invoice number
- User shall filter sales by payment method (Cash/Card/UPI/Wallet/Net Banking)
- User shall filter sales by payment status (Pending/Completed/Refunded)
- User shall filter sales by customer name
- User shall filter sales by date range (created_after/before)
- User shall filter sales by discount range (min/max)
- User shall filter sales by total amount range (min/max)
- User shall filter sales by product within items
- Admin/Manager shall create new sales with line items
- Sale creation shall auto-calculate subtotal, apply discount, calculate total
- Sale creation shall require customer selection
- Sale creation shall allow multiple product line items per sale
- Sale creation shall support 5 payment methods with status tracking
- Sale creation shall auto-generate invoice number (unique)
- Sale creation shall reduce product stock quantity automatically
- Sale creation shall create stock movement log entry (type: OUT, reason: SALE)
- Sale detail page shall display complete transaction details with items
- Sale detail page shall include customer and staff information
- Sale shall support PDF invoice download

- Admin/Manager shall be able to edit sale status (payment_status)
- Sales data shall be exportable to CSV format
- CSV export shall include all relevant sale fields and line items

5. Customer Management

- User shall view paginated list of all customers
- Customer list shall support infinite scroll pagination
- User shall search customers by name, email, phone, or address
- User shall filter customers by date range (created_after/before)
- Admin/Manager shall create new customer records
- Admin/Manager shall edit existing customer information
- Admin/Manager shall delete customer records
- Customer detail page shall display all contact information
- Customer detail page shall show complete order/sales history
- Customer detail page shall calculate total customer lifetime value

6. Supplier Management

- User shall view paginated list of all suppliers
- User shall search suppliers by name or email
- User shall sort suppliers by creation date
- Admin/Manager shall create new supplier records
- Admin/Manager shall edit existing supplier information
- Admin/Manager shall delete supplier records
- Supplier detail page shall display all contact information
- Supplier detail page shall show all linked products
- System shall prevent deletion of suppliers with linked products

7. Stock Movement & Inventory

- System shall create stock movement entry for every inventory change
- Stock movement shall track movement type (IN/OUT)
- Stock movement shall track reason
(Purchase/Sale/Return/Adjustment/Transfer)

- Stock movement shall attribute action to logged-in user
- Stock movement shall include timestamp and optional notes
- User shall view stock movement log with pagination
- User shall search stock movement by product name
- User shall filter stock movement by movement type
- User shall filter stock movement by reason
- User shall filter stock movement by user/username
- User shall filter stock movement by date range
- Stock movement log shall be exportable to CSV
- System shall calculate low stock alerts based on low_stock_limit
- Products with stock below limit shall be highlighted on dashboard

8. Category Management

- User shall view list of all product categories
- Admin/Manager shall create new categories
- Admin/Manager shall edit existing categories
- Admin/Manager shall delete categories
- Category detail page shall show all linked products
- System shall prevent deletion of categories with linked products

9. User Management (Admin Only)

- Admin shall view paginated list of all users
- Admin shall create new users with role assignment (Admin/Manager/Staff)
- Admin shall view user details including role and account status
- Admin shall edit user information and role
- Admin shall delete user accounts
- System shall track user creation and last update timestamps
- Non-admin users shall NOT access user management page

10. Data Export

- User shall export sales data to CSV format
- User shall export stock movement data to CSV format
- CSV export shall include all relevant columns with data

- CSV export shall respect applied filters and date ranges
- System shall generate PDF invoices for sales transactions
- PDF invoice shall include invoice number, items, totals, and payment info

Non functional requirements

Category	Details
Performance	Page load time should not exceed 3 seconds
	List pagination should complete within 1 second
	Search results should be displayed within 500ms
	Chart rendering with 1000+ data points shall complete within 2 seconds
	Infinite scroll should load next batch within 800ms
Scalability	System should support 50+ concurrent user requests
	API should support up to 1000 requests/minute per user
Security	All sensitive data should be transmitted over HTTPS
	JWT tokens should be stored in HttpOnly cookies (not localStorage)
	Passwords should be hashed on backend (never stored in plain text)
	CSRF protection should be implemented for state-changing operations
	Role-based access control shall be enforced server-side
	Session tokens should expire after 24 hours of inactivity
Reliability	System uptime should be 99.5%
	Data backup should occur daily with recovery capability
	Database transactions should have ACID compliance

	Failed API requests should auto-retry up to 3 times
Usability	Frontend should provide consistent navigation (sidebar + route guards)
	Dark mode should be fully functional without UI glitches
	All forms should have clear validation error messages
	Page navigation should work on all modern browsers (Chrome, Firefox, Safari, Edge)
	Keyboard navigation should be fully supported (Tab, Enter, Escape)
Compatibility	Frontend should work on Next.js 15+
	Backend API should be Django REST Framework compliant
	Browser support: Chrome, Firefox, Safari, Edge
Accessibility	Sidebar/dialog controls should include sr-only labels where visual text is hidden.
	Loading indicators should expose semantic role/label.
	Keyboard navigation should support sidebar toggle shortcut.

User flow

Sign In Flow

1. User opens the application.
2. Enters email and password.
3. System validates credentials.
4. System sets authentication cookie on success.
5. User is redirected to the dashboard.

Create Sale Flow

1. Staff or authorized user opens the sales screen.
2. Selects a customer and adds sale items.
3. System calculates totals and validates stock.
4. User submits the sale.
5. Backend saves the sale and sale items.
6. Background jobs generate invoice PDF and send email if required.
7. User can download the invoice from the sale detail view.

Password Reset Flow

1. User requests a password reset by email.
2. System generates a 6-digit OTP and sends it via email.
3. User submits OTP with a new password.

4. System validates OTP and password policy.
5. System updates the password and removes the OTP record.

Low-Stock Alert Flow

1. Sale or stock movement updates product quantity.
2. System checks whether inventory is below threshold.
3. If threshold is crossed, background task sends alert email to admin users.

Acceptance Criteria

Authentication

- Valid credentials return a successful login response and set a secure authentication cookie.
- Invalid credentials must return a failure response without creating a session.
- Protected endpoints must reject unauthenticated requests.

Password Reset

- The system must send a 6-digit OTP to the registered email address.
- OTPs must expire after the configured validity period.
- Password reset must fail if OTP is invalid, expired, or missing.

Inventory and Sales

- Authorized users must be able to create, edit, and delete inventory master data.
- Creating a sale must persist the sale and its line items.
- Sale detail must support invoice PDF download.
- Low-stock conditions must trigger an alert task.

Dashboard and Reporting

- Dashboard endpoints must return summary metrics for the selected period.
- CSV export endpoints must return downloadable files with report data.

Caching and Health

- Repeated reads on list endpoints should be served from cache when applicable.
- Writes must invalidate the relevant cache keys or patterns.
- Health check must return success only when database and Redis are reachable.

Assumptions

- The backend runs on Django REST Framework and the frontend on Next.js 15.
- PostgreSQL and Redis are available in the deployed environment.
- SMTP credentials are configured for outbound email.
- Users access the application through a modern web browser.
- Role rules are enforced consistently in both backend and frontend.

Out of Scope

- Payment gateway integration (Stripe, PayPal, etc.)
- Barcode/QR code scanning
- Advanced BI and custom dashboards
- Multi-tenant architectures
- Mobile native apps
- Offline-first synchronization
- Audit trail / change logs beyond created/updated timestamps

Success Metrics

- Login success rate above 99% for valid credentials.
- Invoice generation and delivery initiated without blocking the main API response.
- Cache hit rate improves response time on repeat list reads.
- Low-stock and OTP emails are delivered within expected operational time.

- Users can complete common sales tasks without support intervention.

Sample Test Scenarios

TC ID	Scenario	Precondition	Steps	Expected Result
TC-AUTH-01	Login with valid credentials	User account exists in the system	1. Submit valid username and password 2. Click Sign In	User is authenticated and redirected to /dashboard. Auth cookie is created.
TC-AUTH-02	Login with invalid credentials	None	1. Submit incorrect username or password 2. Click Sign In	Login rejected. Error message shown. User stays on /sign-in. No cookie created.
TC-AUTH-03	Request password reset via OTP	Registered email exists in the system	1. Navigate to /forgot-password 2. Submit the registered email address	OTP dispatched to the email. Success message shown on screen.
TC-AUTH-04	Reset password with valid OTP	Valid OTP received via email	1. Submit OTP and new password 2. Confirm new password field matches	Password updated. OTP invalidated. Login succeeds with new password.

TC-AUTH-05	Logout clears session	User is logged in	1. Click Logout button 2. Check DevTools → Application → Cookies 3. Try navigating to /dashboard manually	Redirected to /sign-in. Auth cookie removed. /dashboard inaccessible after logout.
TC-RBAC-06	Admin has full access to all pages	Logged in as Admin role user	1. Navigate to /users 2. Navigate to /products 3. Check for Create, Edit, Delete buttons on each page	All pages accessible. Full CRUD buttons visible on every module.
TC-PROD-07	Create product with all required fields	Logged in as Admin or Manager. Category and supplier exist.	1. Fill all required fields including SKU, price, and stock quantity 2. Submit the form	Product created and visible in the product list. All fields saved correctly.
TC-PROD-08	Create product with missing required field	Logged in as Admin or Manager	1. Leave the SKU field empty 2. Submit the form	Form submission blocked. Validation error shown on the empty field.
TC-PROD-09	Search products by name	At least 5 products exist in the catalog	1. Type a product name in the search box 2. Clear and search with a term	Matching products returned. Empty state shown when no match found.

			that matches nothing	
TC-PROD-10	Search products by SKU	At least 5 products exist in the catalog	1. Type a valid SKU in the search box	Correct product returned matching that SKU.
TC-PROD-11	Filter products by category	Products with different categories exist	1. Select a category from the filter dropdown	Only products belonging to the selected category are shown.
TC-SALES-12	Create sale — subtotal, discount, and total calculation	Products and a customer exist. Stock is sufficient.	1. Select a customer 2. Add 2 products with quantities 3. Apply a discount amount 4. Submit the sale	Sale saved. Subtotal, discount, and total are mathematically correct. Unique invoice number auto-generated.
TC-SALES-13	Download PDF invoice after sale	A completed sale exists with known line items and totals	1. Open the sale detail page 2. Click Download Invoice 3. Open the downloaded PDF	PDF downloads. Contains invoice number, customer name, line items, totals, and payment method.
TC-SALES-14	Filter sales by date range	Sales data exists across multiple months	1. Apply date filter: Jan 1 – Mar 31 2026 2. Check the results	Only sales within Jan–Mar 2026 are shown.

TC-CUST-15	Search customer by name, email, and phone	At least 3 customers exist	1. Search by name 2. Clear and search by email 3. Clear and search by phone number	Correct customer returned for each search type.
TC-DASH-16	Top 5 products ranked correctly on dashboard	Multiple products have been sold with different revenue totals	1. Navigate to /dashboard 2. Check the Top Products section	Products are ranked by revenue in descending order. Revenue and quantity values are correct.
TC-EXPO RT-17	CSV export — row count matches filtered UI	Sales data exists for multiple months	1. Apply date filter: Jan 1 – Mar 31 2026 2. Note the row count on screen 3. Export as CSV and count rows in the file	CSV row count matches the UI exactly. All expected column headers are present.
TC-EXPO RT-18	PDF invoice contains all required fields	A completed sale exists with known data	1. Open sale detail page 2. Click Download Invoice 3. Open the PDF and verify each field	PDF contains invoice number, customer name, line items, unit prices, subtotal, discount, total, and payment method.